It’s Week 10

How does everyone feel about the exam?

You will learn about

- The format of the final exam
  - How to approach various questions
  - How to frame your thinking
  - How to read a research paper

Exam is open for 3 Hours (on Inspera – remote)
25th August 2025 1pm – 4pm

COMP6452 – Software Architecture for Blockchain Applications

Final Exam – Term 2 2025

INSTRUCTIONS:

1. Time allowed – 3 hours including reading time.
2. Total number of questions to be answered – 14 : 9 (research paper analysis) + 5 (short answers)
3. Total marks available – 35 marks, worth 35% of the total marks for the course.
4. Marks available for each question are shown in the exam.
5. Students are advised to read all examination questions before attempting to answer the questions.
6. Note that each question has a strict word limit.
7. This exam cannot be copied, forwarded, or shared in any way.
8. Students have been reminded of the UNSW rules regarding Academic Integrity and Plagiarism, and the fit to sit policy, before starting the exam.
9. You must complete and upload all your work within the exam time. There is no extra time. No late submissions will be accepted.
10. Your work will be saved periodically throughout the exam and will be automatically submitted when the test ends provided you are connected to the internet. If you are not connected to the Internet at the conclusion of the exam, please download a submission and email it to cs6452@cse.unsw.edu.au.

Exam Details – 35 Marks

There are two parts for the final examination

1. Reviewing a research paper
    - Total questions: 9
    - Maximum marks: 20
2. Questions that require short answers
    - Similar to your quizzes covering the whole syllabus
    - Total questions: 5
    - Maximum marks: 15

Exam Details – Research paper

1. You are required to read the following scientific paper and write a commentary on it based on the instructions below.
You will then submit the commentary as responses to a set of questions.

A Platform Architecture for Multi-Tenant Blockchain-Based Systems

- Note that each question has a **strict word limit**.
- **Do not** copy-paste material from the paper or any other source, instead write an original piece.
- The usual checks for plagiarism will be performed.

I confirm that I managed to successfully download and open the PDF file.

- Yes
- No – Please contact the UNSW Exams Team at exams@unsw.edu.au Note that support for exams is only actively monitored/attended to between 8am - 6pm AEDT.

How to analyse a research paper

https://www.youtube.com/watch?v=Gv5ku0eoY6k

Paper Title:

A Platform Architecture for Multi-Tenant Blockchain-Based Systems

Ingo Weber, Qinghua Lu, An Binh Tran
Data61, CSIRO, Sydney, Australia
firstname.lastname@data61.csiro.au

Amit Deshmukh, Marek Gorski, Markus Strazds
Laava ID Pty Ltd, Sydney, Australia
firstname@laava.id

***Abstract*—Blockchain has attracted a broad range of interests from start-ups, enterprises and governments to build next generation applications in a decentralized manner. Similar to cloud platforms, a single blockchain-based system may need to serve multiple tenants simultaneously. However, design of multi-tenant blockchain-based systems is challenging to architects in terms of data and performance isolation, as well as scalability. First, tenants must not be able to read other tenants' data and tenants with potentially higher workload should not affect read/write performance of other tenants. Second, multi-tenant blockchain-based systems usually require both scalability for each individual tenant and scalability with number of tenants. Therefore, in this paper, we propose a scalable platform architecture for multi-tenant blockchain-based systems to ensure data integrity while maintaining data privacy and performance isolation. In the proposed architecture, each tenant has an individual permissioned blockchain to maintain their own data and smart contracts. All tenant chains are anchored into a main chain, in a way that minimizes cost and load overheads. The proposed architecture has been implemented in a proof-of-concept prototype with our industry partner, Laava ID Pty Ltd (Laava). We evaluate our proposal in a three-fold way: fulfilment of the identified requirements, qualitative comparison with design alternatives, and quantitative analysis. The evaluation results show that the proposed architecture can achieve data integrity, performance isolation, data privacy, configuration flexibility, availability, cost efficiency and scalability.**

***Index Terms*—software architecture, blockchain, smart contract, multi-tenant, Merkle tree**

First Paragraph

I. Introduction

Blockchain is an emerging distributed ledger technology which has attracted a broad range of interests from start-ups, enterprises and governments [14] [19] to address lack-of-trust issues in a decentralized manner. A large number of projects have been conducted to explore how to use blockchain to re-architect systems and to build new applications and business models. Blockchain application areas are diverse, including supply chain, IoT, physical or digital asset registries, digital currency, payment, trade finance, and identity management.

Similar to cloud platforms, a single blockchain-based system is often required to serve multiple tenants who reside in the same system to maintain their data. For example, a traceability system usually provides quality tracking services to different product manufacturers and each manufacturer can manage the tracking of their products individually.

Last Paragraph

We implement the proposed architecture in a proof-of-concept prototype, as part of a collaborative project with our industry partner Laava[1]. We adopt Ethereum [1] in our implementation since it currently offers the most mature smart contract support. In our three-pronged evaluation, we first examine the architecture by checking the fulfilment of the identified requirements for multi-tenant blockchain-based systems. Second, we provide a qualitative analysis by comparing the proposed architecture with two architecture design alternatives. Third, we conduct a quantitative analysis by measuring the throughput under a number of conditions. The results show that the proposed architecture can achieve data integrity, performance isolation, data privacy, configuration flexibility, availability, cost efficiency and scalability.

Recording key headings/subheadings

Key headings:

- Requirements
- Architecture
- Implementation
- Evaluation
- Conclusion

A. Functional Requirements

FR1 – Writing data on blockchain restricted to selected clients: The platform shall have the ability to write data on blockchain, restricted, e.g., to the platform owner.

FR2 – Writing batches of data on blockchain: The platform shall have the ability to write batches of data with low cost and overhead.

FR3 – Viewing the entire history: The platform shall have ability to read the entire data history, i.e., all historical events and data values over time.

FR4 – Tracking authenticity of data: end users need to be able to see and validate the identity of clients that wrote data to the system.

FR5 – Providing external auditing/verification for independent agencies: independent agencies need to be able to access data for auditing for each individual tenant.

FR6 – Providing a multi-tenant platform: the platform supports multiple tenants to serve their end users, where different tenants can have different business needs.

B. Non-Functional Requirements

NFR1 – Data integrity of on-chain data must be ensured.

NFR2 – Scalability:

- Scalability within each tenant. For example, a tenant might store large amounts of data within a period of time.
- Scalability in the number of tenants.

NFR3 – Data Privacy: in general, tenants must not be able to read other tenants' data (e.g., how many unique item IDs were created, scan event counts, timing, or locations).

NFR4 – Performance Isolation: tenants with potentially higher workload (e.g., commodity goods with millions of events daily) should not affect read/write performance for other tenants.

NFR5 – Availability: the blockchain infrastructure must be available, in terms of responsiveness to read/write operations.

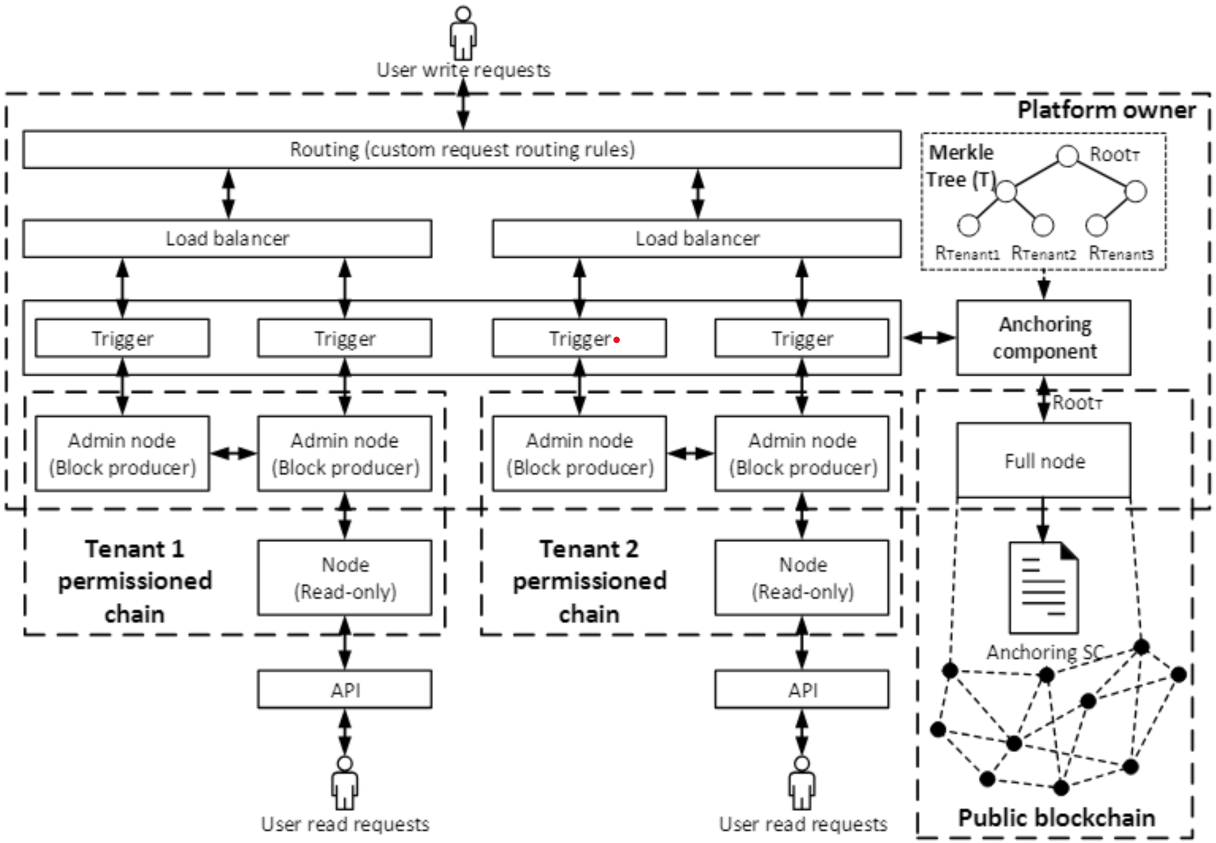


Figure 2. Platform Architecture for multi-tenant blockchain-based systems.

A. Overall Architecture

Fig. 2 illustrates the platform architecture we propose for multi-tenant blockchain-based systems. Each tenant has an individual permissioned blockchain to maintain their information. The platform owner hosts all admin nodes for producing blocks while the tenants/auditors host read-only nodes, which can be enforced through the blockchain configuration. The platform owner has access to all the tenants' on-chain data, and provides APIs for writing to tenants' chains.

An end user, e.g., a client of a tenant, sends write requests through the routing component, which forwards them to respective tenant's blockchain to process. The load balancer distributes the workload to the trigger components for different admin nodes of the same tenant. The functionality provided by the trigger includes writing data to the blockchain and communicating with the anchoring component. The user sends read requests to the read-only node through a public API.

The proposed architecture allows publicly verifiable integrity of private blockchains at particular times via anchoring the consensus state of each private chain at periodic intervals to the public blockchain. The anchoring component connects to a node in the public blockchain network, and one node for each tenant's blockchain to be anchored. We design a Merkle tree T, as shown in the top right corner of Fig. 2, in which each leaf node represents the root of each tenant blockchain

Figure 7. Deployment architecture for quantitative analysis experiments.

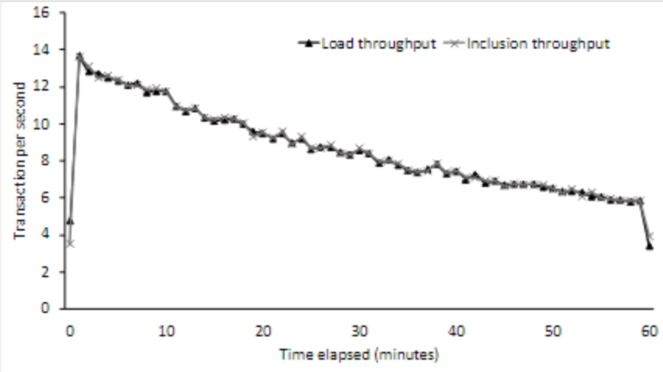


Figure 8. Throughput for Test 1, normal load scenario (<15tps).

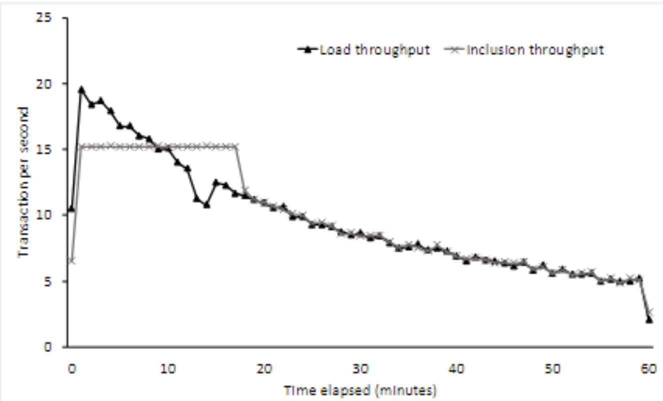


Figure 9. Throughput for Test 2, boundary load scenario (starting at ≈18tps).

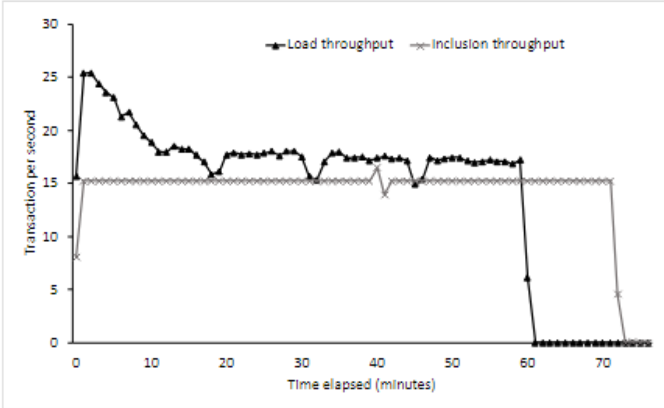


Figure 10. Throughput for Test 3, overload scenario (≈18-25tps).

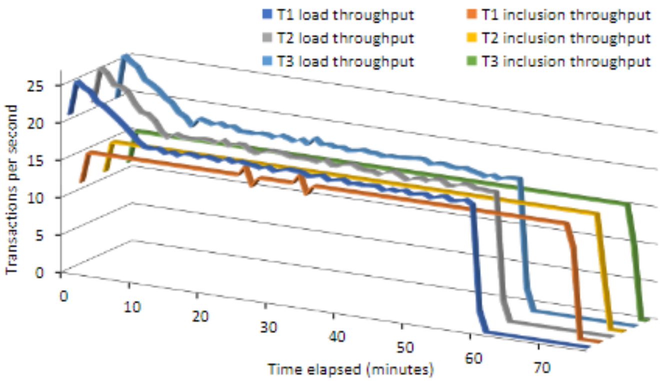


Figure 11. Throughput for Test 4, overload scenario (≈18-25tps).

corresponding to 304 unique ID creations per second. We run four different tests to measure the transaction sending and inclusion throughput over time with different loads:

Test 1: normal load scenario (<15tps), one tenant chain.

Test 2: boundary load scenario (starting at ≈18tps), one tenant chain.

Test 3: overload scenario (≈18-25tps), one tenant chain, i.e., the incoming throughput is higher than the theoretical throughput limit of the blockchain.

Test 4: an overload scenario with three tenant chains (≈18-25tps on each chain), i.e., Test 3 on three tenant chains in parallel. With this test we investigate the performance isolation between tenant chains as well as the anchoring protocol.

VII. Conclusion and Future Work

This paper presents a platform architecture for multi-tenant blockchain systems. In the design, each tenant is given an individual blockchain, and all tenant chains are anchored to a public blockchain periodically. The anchoring uses a combined root of all tenant chains, thus achieving data integrity, low cost, and performance and data isolation. The proposed architecture has been implemented in a prototype with our industry partner, Laava. We evaluate the solution in a three-pronged fashion: by examining requirement fulfilment, by quantitative comparison with two design alternatives, and by quantitative analysis using the prototype. The system achieves all objectives.

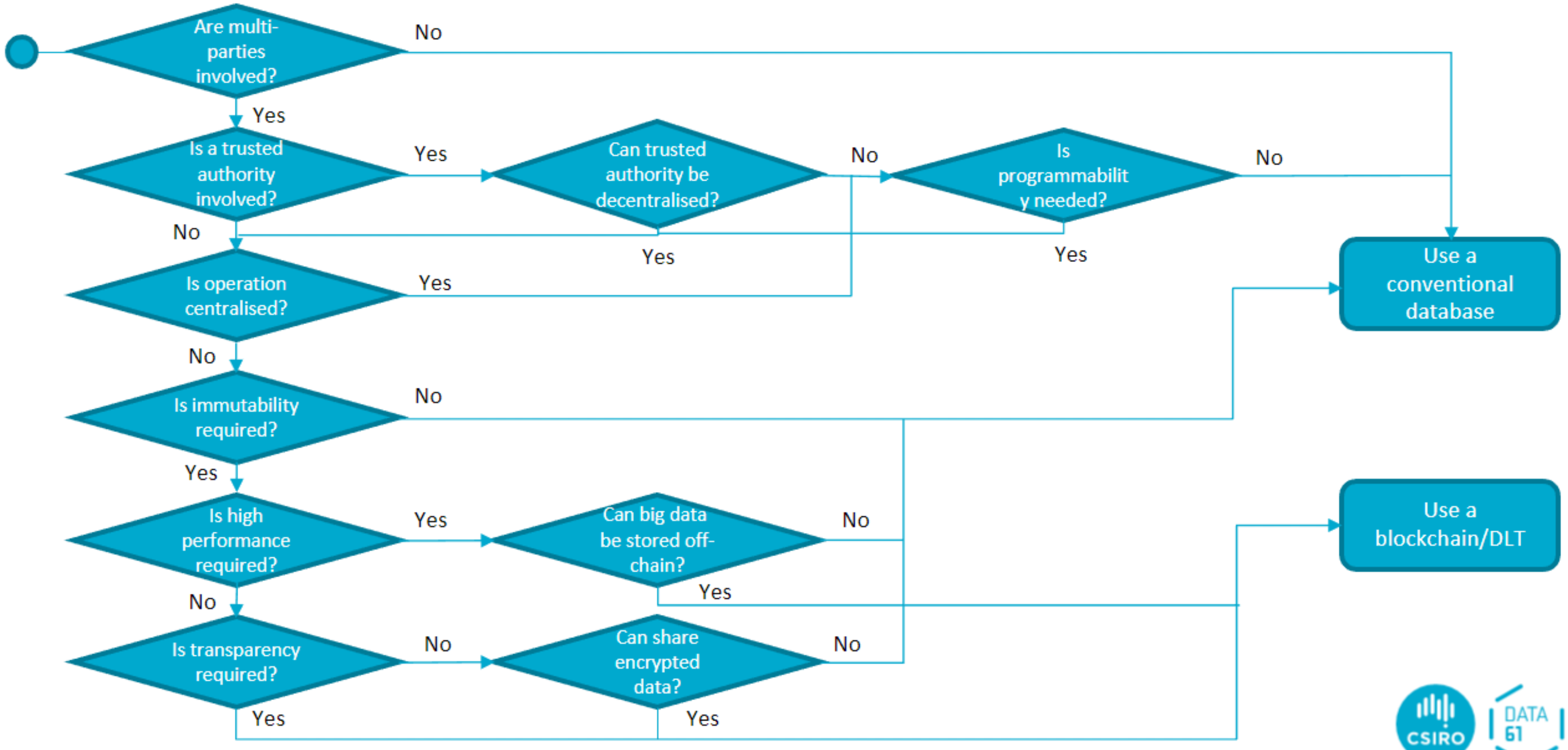
Although we focused on multi-tenant blockchain-based systems, the proposed architecture can be applied to many situations requiring multiple blockchains. Examples include a long-lived and a short-lived blockchain for long and short-running business needs, or a separate blockchain per year.

In future work, we plan to explore the above-mentioned flexible use of anchored chains, as well as the use of other technology platforms – both for tenant chains and public chains – with a single anchoring component.

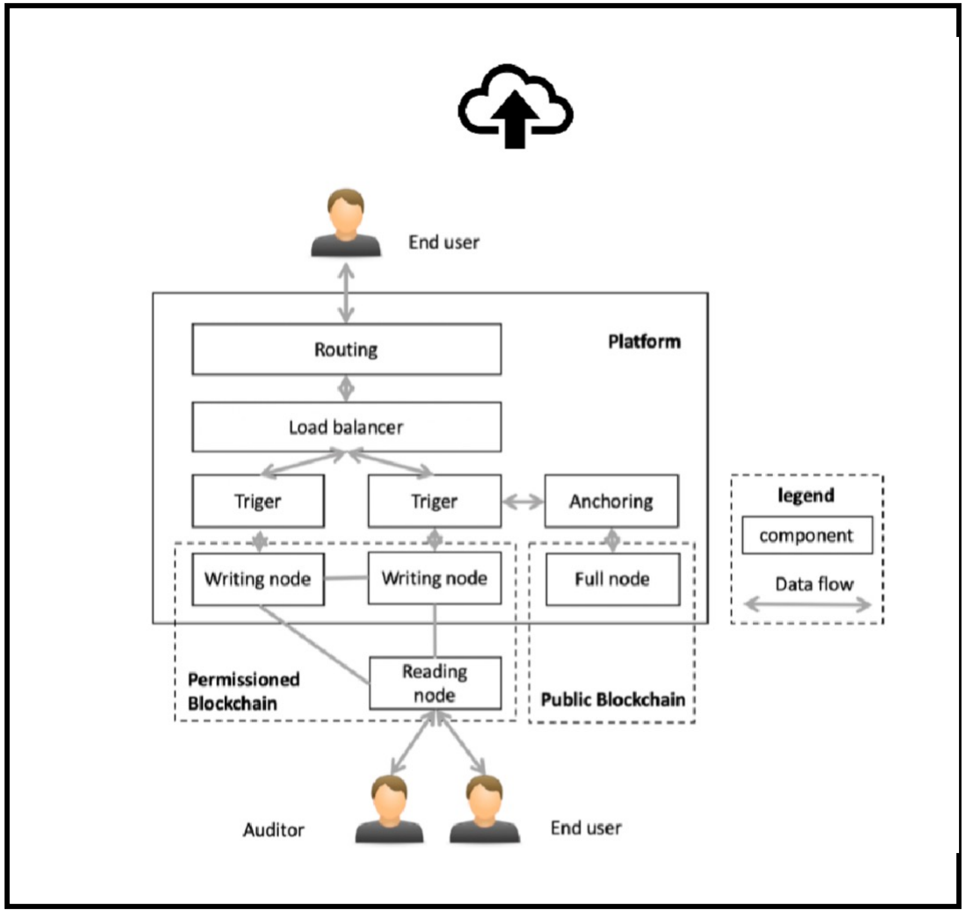
A Sample Research Paper and Possible Answers

Go through the sample research paper and the sample answers provided.

Blockchain Suitability Assessment (One of the Ways)



Possible Architecture Diagram



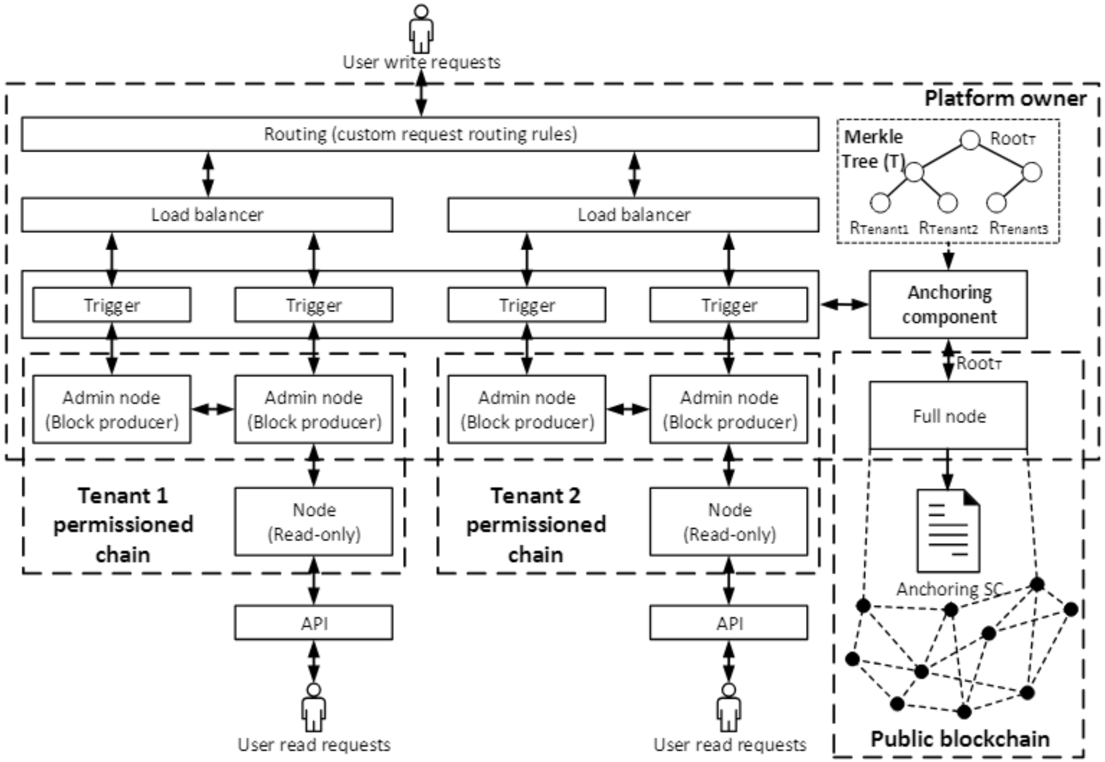
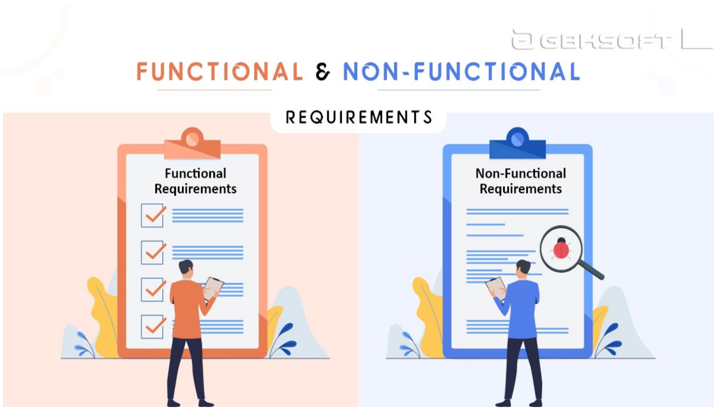


Figure 2. Platform Architecture for multi-tenant blockchain-based systems.

Critiquing a Blockchain Design

Design Patterns

Requirements



Trilemma

Best wishes for the Final Exam